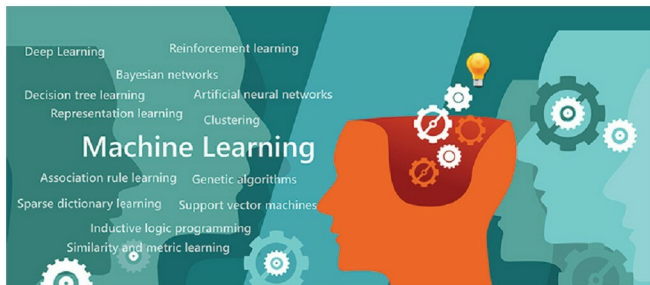


Machine Learning Basics for Newbies



Machine learning focuses on the development of computer programs that can teach themselves to grow and adapt when exposed to new data. It is increasingly impacting our lives nowadays, as machines play an important role in banking and financial services, healthcare, retail, publishing, and in the social media, robot locomotion and gaming domains.

When we start the journey of life as new born babies, we inherit the characteristics of our parents. We don't know what to do and when to do what. As we grow up, our parents and elders teach us how to walk, talk and take various decisions in our lives and, as time passes, we gain experience and knowledge. Finally, we start taking our own decisions based on our learning and experience. Similarly, when we write any code to make a system do any work, the system only does what we ask it to do—it cannot think or take any extra decisions on its own nor perform actions on that basis. Yet, machine learning actually teaches the system to learn and take decisions when exposed to a new set of data on the basis of the experience it gains while performing different actions. Nowadays, it is an emerging technology that is widely being implemented across all types of industries. Google's self-driving cars, flying drones, anomaly detection or Big Data processing are among the recent examples of this technology being used.

Machine learning is one type of artificial intelligence (AI) that provides computers with the ability to learn without being explicitly programmed. It uses pattern recognition and computational learning theory to study and develop algorithms (which can learn from the sets of available data), on the basis of which it makes decisions. These algorithms work by building a model (such as the Predictive Model or Neural Network Model) from sample inputs in order to make data-driven decisions. These models help in developing

Decision Trees, using which, the system makes its decision.

Machine learning makes use of mathematical optimisation to deliver different theories, methods and application domains for a specific field. It uses the data mining technique to perform exploratory data analysis over a set of data in order to make predictions. This is basically referred to as unsupervised learning. Machine learning helps data scientists, engineers, researchers and analysts to take a reliable decision by uncovering the hidden insights acquired through the analysis of historical trends in the data.

Types of machine learning

The tasks performed using machine learning are classified broadly into three categories, based on the nature of the learning signal available to a learning system (the system which helps to make decisions).

1. **Supervised learning:** This is a type of machine learning in which the system is presented with a set of labelled training data (inputs and their corresponding set of outputs). Now, it is the task of supervised machine learning to predict a new set of outputs for a given new set of inputs by learning or finding out a general rule or pattern that maps the given set of inputs to their corresponding outputs. The pattern or rule that helps in predicting output is generally denoted by a specific function. Supervised learning is further classified as regression and classification problems, on the basis of the methodology that is implemented to find a specific pattern.